Bus Route Management Using Circular Linked List

1. Problem Statement

In a circular bus route, the bus travels through a fixed sequence of stops and returns to the starting stop after reaching the last stop. The bus stops need to be represented efficiently such that the last stop is connected back to the first stop. Implement a circular linked list to represent and display the bus route.

2. Solution

A circular linked list is used to represent the bus stops. Each node stores the name of a bus stop and a pointer to the next stop. The next pointer of the last node points back to the first node, forming a circular route.

The program performs the following operations:

- Create a node for each bus stop.
- Insert the new stop at the end of the route.
- Connect the last stop back to the first stop.
- Traverse and display all bus stops until the starting stop is reached again.

3. Objectives – Bus Route Management

- To represent bus stops using a circular linked list.
- To implement insertion of new bus stops in the circular route.
- To display all bus stops by performing circular traversal.
- To understand the use of nodes and pointers in a circular linked list.
- To demonstrate how the last bus stop connects back to the first stop.
- To understand the practical application of a circular linked list in bus route management.